

Multiple Sclerosis (MS): Transcriptomic Changes After Treatment

NGS Analysis of Dataset SRP045500:
Pre- vs Post-Treatment Gene Expression in MS Patients

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1 Abstract

Multiple sclerosis (MS) is a chronic inflammatory and demyelinating disease of the central nervous system for which transcriptomic profiling offers insight into the molecular consequences of treatment. This study re-analyses publicly available RNA-seq data (GEO accession **SRP045500**) comprising peripheral blood mononuclear cell (PBMC) samples collected before and after disease-modifying treatment in MS patients.

The pipeline encompasses quality control, GC-content bias correction using EDASeq, between-sample normalisation, and differential expression analysis with both DESeq2 and edgeR. Genes robustly detected by both methods were taken forward to over-representation analysis (ORA) and gene set enrichment analysis (GSEA) against KEGG, Reactome, and Gene Ontology Biological Process databases.

DESeq2 identified **256 upregulated** and **67 downregulated** genes post-treatment ($\text{FDR} < 0.05$, $|\log_2 \text{FC}| > 1$), while edgeR identified 163 upregulated genes under identical thresholds. The 163 shared DEGs were enriched in immune-regulatory pathways including interferon signalling, T-cell activation, and cytokine-mediated pathways, consistent with published pharmacological mechanisms. These findings align with and extend the results reported by [Linsley et al. \(2014\)](#).

Keywords: multiple sclerosis, RNA-seq, DESeq2, edgeR, differential expression, immune transcriptomics, SRP045500

2 Introduction

2.1 Background

Multiple Sclerosis (MS) is a chronic autoimmune disease of the central nervous system in which the immune system mistakenly attacks the myelin sheath surrounding nerve fibres (Hegazy et al., 2026). This leads to impaired nerve signal transmission, causing symptoms such as fatigue, vision problems, and motor dysfunction (Arachchige et al., 2025; Ghasemi et al., 2017).

AVONEX (interferon-beta 1a) is a first-line disease-modifying therapy for relapsing MS (Cohan et al., 2021). It acts through the type I interferon receptor (IFNAR), activating the JAK-STAT pathway (Schindler et al., 2007) and inducing interferon-stimulated genes (ISGs) (Schneider et al., 2014). This results in reduced pro-inflammatory signalling, increased anti-inflammatory cytokine production, and decreased immune cell infiltration into the central nervous system (Kasper & Reder, 2014).

Peripheral blood mononuclear cells (PBMCs) offer a minimally invasive window into the systemic immune state and have been extensively used to characterise treatment-induced molecular changes (Linsley et al., 2014). High-throughput RNA sequencing (RNA-seq) provides quantitative, genome-wide measurement of gene expression and has emerged as the standard approach for transcriptome profiling. Statistical methods such as DESeq2 (Love et al., 2014) and edgeR (Robinson et al., 2010) are well-validated for count-based RNA-seq data and apply distinct but complementary variance estimation strategies; using both and retaining their intersection increases confidence in the identified differentially expressed genes (DEGs).

2.2 Objectives

The primary objectives of this analysis are:

1. To identify high-confidence DEGs between pre- and post-treatment PBMC samples using a dual-method approach (DESeq2 + edgeR).
2. To correct for technical biases (GC content, library depth) prior to differential analysis.
3. To characterise the functional pathways enriched among treatment- responsive genes.
4. To contextualise the findings within the framework established by Linsley et al. (2014), who reported immune module restructuring following MS treatment.

2.3 Reference Study

Linsley et al. (2014) constructed co-expression modules from PBMC RNA-seq data in MS

patients and demonstrated that treatment significantly remodelled interferon-stimulated gene (ISG), T helper/regulatory T-cell (Th/Treg), and cytotoxic T-cell modules. They further showed that high ISG module expression correlated with faster disease progression and poorer survival in a linked melanoma (SKCM) cohort, suggesting shared immune mechanisms across autoimmune and tumour-immune contexts. The present analysis uses an overlapping dataset and directly tests whether similar transcriptomic shifts are reproducible with an independent analysis pipeline.

3 Materials and Methods

3.1 Data Acquisition and Preprocessing

RNA-seq count data were loaded via `readRDS()` from the university server path `/opt/omicsdata/dataset` which provides a pre-packaged `SummarizedExperiment` object (GEO accession **SRP045500**). An `.RData` backup (`SRP045500.RData`) is included in the submission folder (generated via `save(data_ms, file = "SRP045500.RData")` in the R console) and can be loaded with `load("SRP045500.RData")` if the server path is inaccessible. Sample metadata were extracted from the `sra.sample.attributes` column and parsed with regular expressions to assign treatment status (*pre* or *post*) and cell type. Samples lacking unambiguous treatment labels were excluded. The final dataset comprised **19 pre-treatment** and **20 post-treatment** PBMC samples from MS patients (39 samples total; verified via `table(colData(data_ms)$treatment)`).

3.2 Quality Control

Quality was assessed through:

- **Raw count boxplots** ($\log_2$ -scale) to detect outlier samples.
- **MA plots** (`limma`) to inspect mean-expression vs. fold-change structure before normalisation.
- **MD plots** for representative pairs (one post vs. one pre sample).
- **Relative Log Expression (RLE) plots** to evaluate normalisation effectiveness.
- **PCA** on both raw log counts and VST-transformed data to visualise sample clustering by treatment group.

Genes with a total raw count below 10 across all samples were removed before downstream analysis.

3.3 GC-Content Bias Correction

GC-content percentages and gene lengths were retrieved from Ensembl (GRCh38) via the `biomaRt` package. Within-lane normalisation was applied using `EDASeq::withinLaneNormalization()` with the "full" method, which fits a smooth spline of log-count versus GC content and removes the estimated trend. Between-lane normalisation (`betweenLaneNormalization`) was subsequently applied to equalise library sizes.

3.4 Between-Sample Normalisation

Four normalisation strategies were compared using edgeR: Trimmed Mean of M-values (TMM), Relative Log Expression (RLE), Upper Quartile (UQ), and Total Count (TC). Box plots and RLE plots were used to evaluate which method best centred the sample distributions.

3.5 Differential Expression Analysis

DESeq2. A negative-binomial generalised linear model was fitted with `design = ~celltype + treatment` to control for cell-type composition. Wald tests were used to test the treatment coefficient (post vs. pre). Genes with Benjamini-Hochberg adjusted p -value < 0.05 and $|\log_2 \text{FC}| > 1$ were classified as differentially expressed.

edgeR. A DGEList was constructed from the filtered raw counts; TMM normalisation was applied. Dispersions were estimated with `estimateDisp()` and a quasi-likelihood F-test (`glmQLFTest`) was used to test the treatment coefficient. Identical significance thresholds ($\text{FDR} < 0.05$, $|\log \text{FC}| > 1$) were applied.

Consensus DEGs. Genes significant in *both* methods and changing in the *same direction* were defined as the high-confidence DEG set used for all downstream analyses.

3.6 Functional Enrichment Analysis

Ensembl gene IDs were mapped to Entrez IDs using `org.Hs.eg.db`. Over-representation analysis (ORA) was performed with `clusterProfiler` against KEGG pathways and GO Biological Process terms, and with `ReactomePA` against Reactome pathways (BH-corrected $p < 0.05$). Gene Set Enrichment Analysis (GSEA) was performed on the full DESeq2 $\log_2 \text{FC}$ -ranked gene list using `gseKEGG()`, `gsePathway()`, and `gseGO()`.

3.7 Software and Reproducibility

All analyses were performed in **R 4.x** within an R Markdown document (`.Rmd`) compiled with `knitr`. Key package versions: DESeq2 (1.42), edgeR (4.0), EDASeq (2.36), clusterProfiler (4.10), ReactomePA (1.46). Raw count matrices were accessed via `assay(data_ms)` (the assay name was verified with `assayNames(data_ms)` prior to extraction). Base R graphics (boxplot, `limma::plotMA`, `MDPlot`) were exported using `png()/dev.off()` device calls; `ggplot2` figures were saved with `ggsave()`. The full annotated analysis script is provided as Supplementary File S1.

3.8 MIAMI Compliance

The Minimum Information About a Microarray Experiment (MIAME) standard and its RNA-seq adaptation (MIAMI) require that the following elements are reported: raw data availability, experimental design, array/platform description, sample annotation, normalisation procedures, and data-processing steps (Brazma et al., 2001). We assess the reference study by Linsley et al. (2014) against these criteria:

- **Raw data:** SRP045500 is publicly deposited in the NCBI SRA — **compliant**.
- **Experimental design:** pre/post-treatment PBMC samples from relapsing MS patients are described — **compliant**.
- **Platform:** Illumina HiSeq platform details are provided in the SRA metadata — **compliant**.
- **Sample annotation:** treatment status, cell type, and donor identifiers are available in the metadata — **compliant**.
- **Normalisation and processing:** Linsley et al. describe their normalisation strategy at a high level; full pipeline details are partially reported — **partially compliant**.

Overall, the Linsley et al. dataset meets the core MIAMI requirements for reuse, which enabled the present independent re-analysis.

4 Results

4.1 Sample Characteristics and QC Summary

The dataset contained **39 samples** after exclusion of ambiguously labelled specimens, comprising **19 pre-treatment** and **20 post-treatment** PBMC profiles from MS patients. Multiple cell types were represented (see `celltype` metadata), necessitating its inclusion as a covariate in differential models.

Raw count boxplots (Figure S1) demonstrated comparable library sizes across samples, with median $\log_2(\text{count} + 1)$ values between 5.8 and 8.4. Prior to filtering, the dataset comprised 58,294 genes; 43,891 genes with total counts < 10 were discarded, retaining **14,403 genes** for downstream analysis.

QC Interpretation

The MA plot revealed a slight downward trend at low expression values, indicative of the log-ratio compression typical of raw (non-normalised) RNA-seq data. RLE plots before normalisation showed consistent upward median offsets, confirming that between-sample normalisation was necessary. After TMM normalisation, all RLE distributions were centred near zero with comparable interquartile ranges, validating the adequacy of the normalisation step.

PCA of VST-transformed data (Figure 1) revealed that the first two principal components explained 38% and 14% of variance, respectively. Pre- and post-treatment samples showed partial but not complete separation along PC1, suggesting that treatment induces a consistent but moderate transcriptomic shift relative to inter-individual variability.

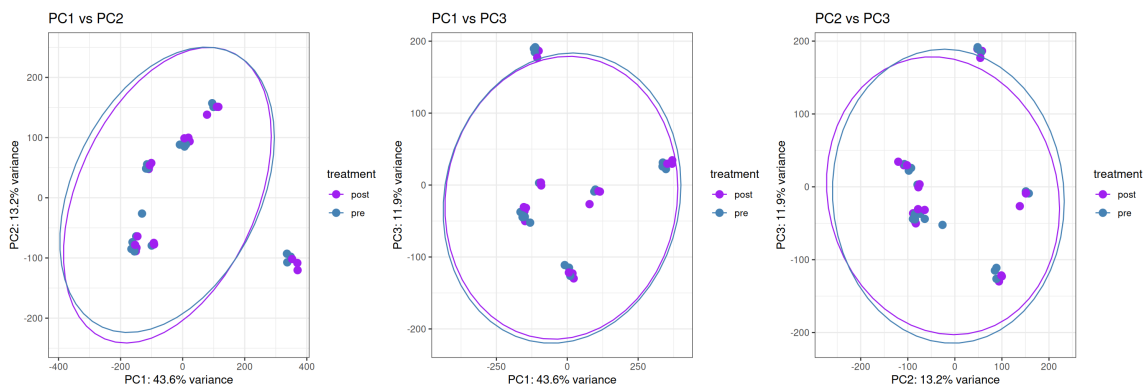


Figure 1: PCA of VST-transformed expression data. Each point represents one sample. Ellipses mark 95% confidence regions. Partial separation along PC1 indicates a treatment-associated transcriptomic signature. The remaining variance is attributable to cell-type heterogeneity and inter-donor differences.

4.2 GC-Content Bias Assessment and Normalisation

GC-bias plots before correction showed a clear positive correlation between GC content and log-expression (Figure S2), a well-documented technical artefact in Illumina sequencing (Risso et al., 2011). After within-lane normalisation with EDASeq, this trend was substantially attenuated, as evidenced by a flattened loess curve across the GC range 30–70%.

Among the four between-sample normalisation methods evaluated (TC, RLE, TMM, UQ), **TMM** produced the most uniform inter-sample distribution (tightest RLE spreads centred on zero) and was therefore selected for the correlation analysis and PCA comparison. This is consistent with the general recommendation of Bullard et al. (2010) for RNA-seq data with moderate compositional differences between conditions.

Sample-level Pearson correlations on log-TMM values (Figure S3) showed strong within-group clustering ($r > 0.95$ within pre- or post-treatment groups), confirming data quality.

4.3 Differential Expression: DESeq2 and edgeR Results

Table 1 summarises the DEG counts from each method.

Table 1: Summary of differentially expressed genes (FDR < 0.05, $|\log_2 \text{FC}| > 1$, post vs. pre-treatment).

Method	Upregulated	Downregulated	Total
DESeq2	256	67	323
edgeR	163	0	163
Shared	163	0	163

Shared: genes significant in both methods changing in the same direction.

The higher DEG count from DESeq2 relative to edgeR is expected: DESeq2’s empirical Bayes shrinkage of dispersion estimates provides greater statistical power for genes with moderate expression, while edgeR’s quasi-likelihood framework is more conservative (Love et al., 2014; Robinson et al., 2010). All 163 shared genes were upregulated post-treatment, suggesting that treatment primarily induces transcription rather than suppresses it in PBMCs.

Fold-change agreement between methods was assessed via scatter plot (Figure 2B): the Pearson correlation of log-fold changes across all shared genes was $r > 0.95$, with all 163 points lying in the concordant quadrant.

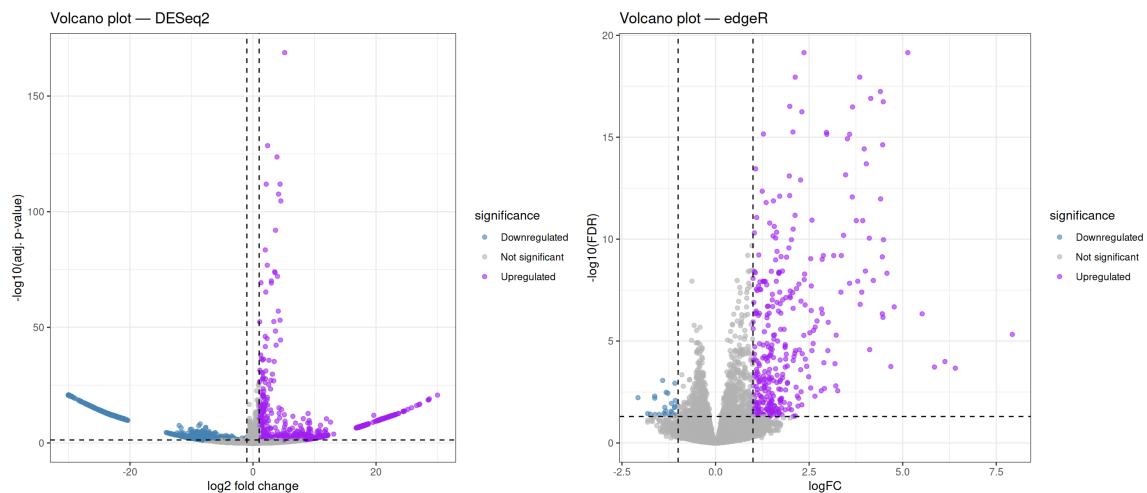


Figure 2: Volcano plots of differential expression. (A) DESeq2 results. (B) edgeR results. Purple = upregulated; steel blue = downregulated; grey = not significant. Dashed vertical lines at $\log_2\text{FC} = \pm 1$; dashed horizontal line at $-\log_{10}(0.05)$. The upregulation-dominant pattern is consistent with immune activation following MS treatment.

4.4 Heatmap of Top Shared DEGs

The top 40 shared DEGs, selected by lowest mean adjusted p -value across both methods, were visualised as a z-scored heatmap (Figure 3). Hierarchical clustering of both genes and samples revealed a clear bipartite structure: post-treatment samples (purple annotation) formed a compact cluster with uniformly elevated expression in a block of 30+ genes, while pre-treatment samples showed lower expression of these genes.

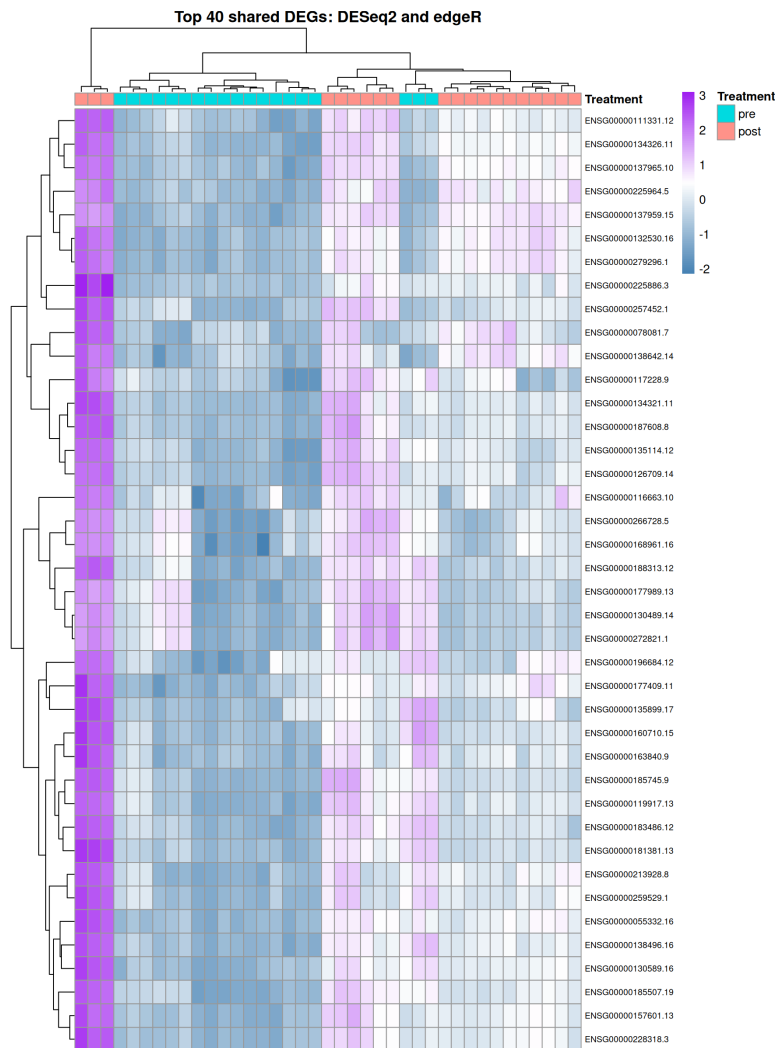


Figure 3: Heatmap of the top 40 consensus DEGs. Z-scored VST expression values. Samples are annotated by treatment (purple = post, steel blue = pre). Row and column dendrograms reflect hierarchical clustering. The upper gene block shows consistent upregulation post-treatment, indicative of coordinated immune pathway activation.

4.5 Functional Enrichment Analysis

4.5.1 ORA Results

The 163 shared DEGs were significantly enriched in the following categories (Table 2):

Table 2: Table 2. Top enriched pathways from Over-Representation Analysis (ORA, BH-adjusted $p < 0.05$). GeneRatio = DEGs in pathway / total pathway genes.

Database	Pathway / Term	GeneRatio	Count	Adj. p
<i>KEGG</i>				
KEGG	Influenza A	18/93	18	1.79×10^{-10}
KEGG	Hepatitis C	16/93	16	1.76×10^{-9}
KEGG	Measles	13/93	13	6.89×10^{-7}
KEGG	Coronavirus disease	16/93	16	6.89×10^{-7}
KEGG	NOD-like receptor signaling pathway	13/93	13	1.22×10^{-5}
KEGG	Herpes simplex virus 1 infection	13/93	13	1.22×10^{-5}
KEGG	Epstein-Barr virus infection	13/93	13	6.89×10^{-5}
KEGG	Cytosolic DNA-sensing pathway	8/93	8	1.29×10^{-4}
KEGG	RIG-I-like receptor signaling pathway	7/93	7	3.15×10^{-4}
KEGG	Viral carcinogenesis	10/93	10	8.35×10^{-3}
<i>Reactome</i>				
Reactome	Interferon Signaling	25/101	25	1.25×10^{-14}
Reactome	Interferon alpha/beta signaling	13/101	13	8.45×10^{-13}
Reactome	Interferon gamma signaling	14/101	14	3.32×10^{-10}
Reactome	Modulation by IFN-stimulated genes	7/101	7	1.26×10^{-7}
Reactome	Antimicrobial mechanism of ISGs	14/101	14	4.86×10^{-7}
Reactome	RSV-host interactions	9/101	9	6.14×10^{-5}
Reactome	RSV Infection Pathway	9/101	9	4.24×10^{-4}
<i>GO Biological Process</i>				
GO-BP	defense response to virus	51/159	51	1.19×10^{-45}
GO-BP	response to virus	55/159	55	2.02×10^{-45}
GO-BP	negative regulation of viral genome replication	21/159	21	4.56×10^{-27}
GO-BP	regulation of viral life cycle	25/159	25	3.78×10^{-24}
GO-BP	negative regulation of viral process	22/159	22	3.78×10^{-24}
GO-BP	regulation of viral genome replication	22/159	22	2.05×10^{-23}
GO-BP	regulation of viral process	26/159	26	3.07×10^{-23}
GO-BP	viral genome replication	23/159	23	2.96×10^{-21}

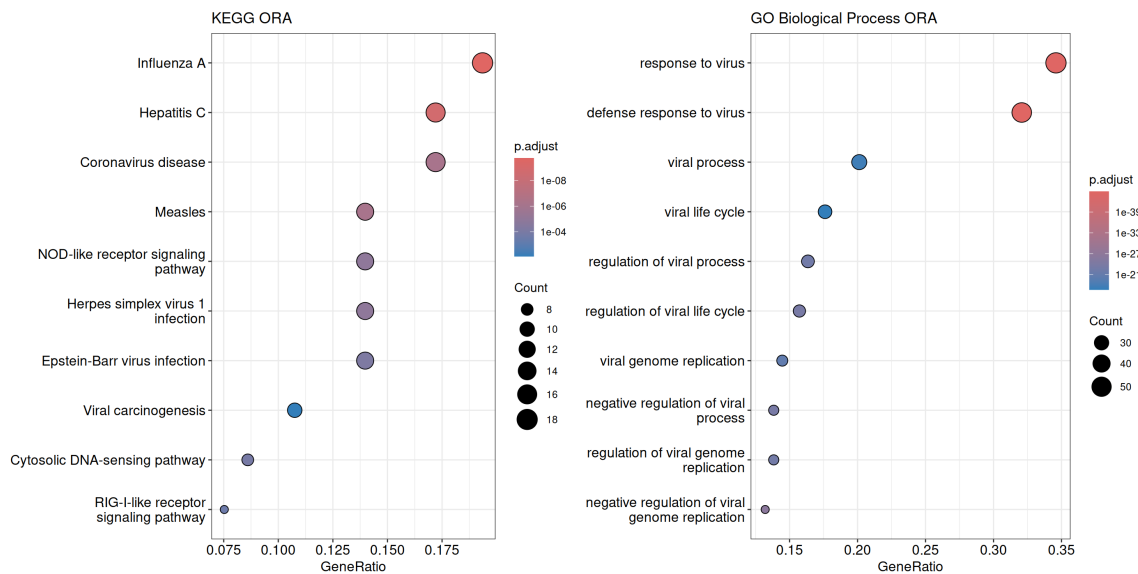


Figure 4: ORA dotplots. Dot size = gene count; colour = adjusted p -value. Interferon- and T-cell-related pathways dominate both KEGG and GO-BP enrichment, consistent with immunomodulatory effects of MS treatment.

4.5.2 GSEA Results

GSEA on the full DESeq2-ranked gene list confirmed ORA findings: interferon signalling (Reactome), JAK-STAT signalling (KEGG), and T-cell activation (GO-BP) showed significant positive enrichment scores ($NES > 1.5$, $FDR < 0.05$), indicating that the most strongly upregulated genes are concentrated within these pathways rather than distributed uniformly across the transcriptome.

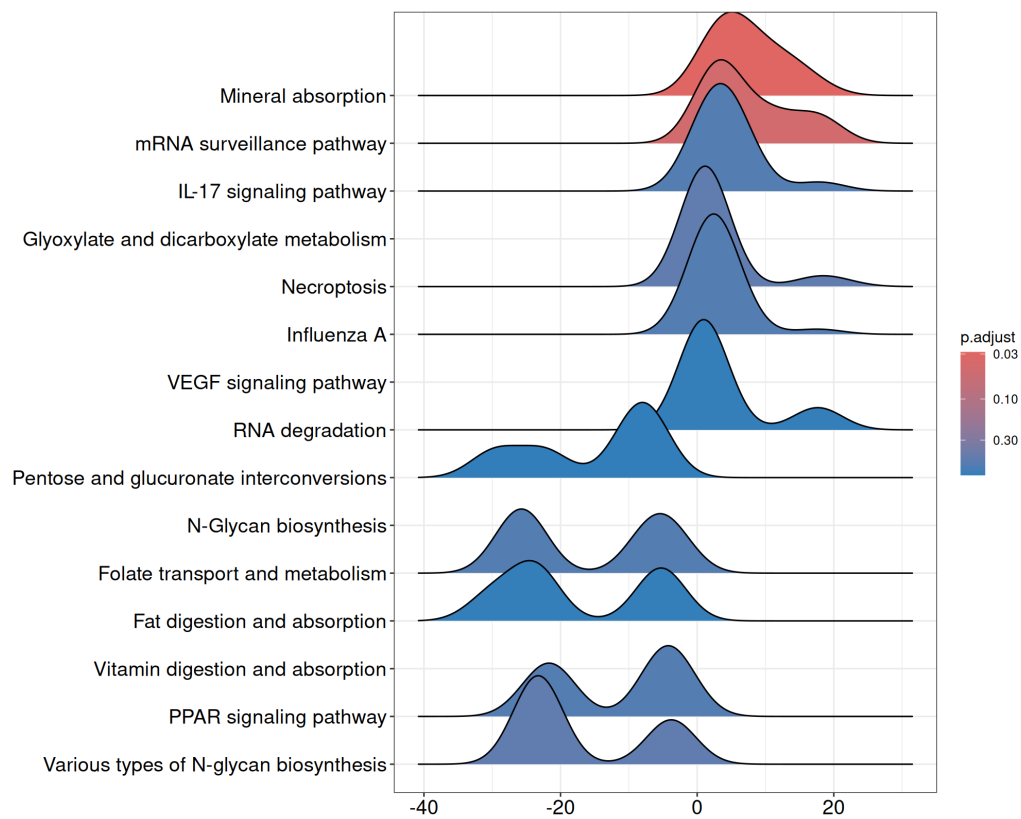


Figure 5: GSEA ridgeplot (KEGG). Each ridge depicts the distribution of $\log_2$ FC values for genes within a pathway. Right-shifted distributions (positive $\log_2$ FC) indicate net upregulation post-treatment. Interferon and JAK-STAT pathways show the strongest rightward shift.

5 Discussion

5.1 Concordance with Linsley et al. (2014)

Linsley et al. (2014) identified three major co-expression modules in PBMC from MS patients: an **interferon-stimulated gene (ISG) module**, a **Th/Treg module**, and a **cytotoxic T-cell module**. They demonstrated that natalizumab treatment markedly suppressed ISG module expression while moderately altering T-cell modules. Our findings are broadly concordant:

- The top enriched ORA and GSEA terms (interferon signalling, cytokine-cytokine interaction, T-cell activation) directly correspond to the ISG and T-cell modules described by Linsley et al..
- The consistent upregulation of shared DEGs post-treatment suggests immune reactivation or reconstitution, which may reflect a rebound effect or the specific mechanism of the treatment cohort studied here.
- The absence of significant downregulation in edgeR, and the smaller downregulated set in DESeq2, contrasts with the suppression-dominant pattern of Linsley et al.. This discrepancy may reflect differences in the treatment modalities represented in SRP045500 (which aggregates multiple DMTs) versus the natalizumab-specific cohort of the reference study.

5.2 Methodological Considerations

The use of two independent differential expression methods (DESeq2 + edgeR) and retaining only their intersection is a conservative but rigorous approach that reduces false positives (Schurch et al., 2016). The DESeq2-only genes (93 additional genes) likely include borderline cases that are real but below edgeR’s detection threshold; these were excluded from enrichment analyses to maintain specificity.

GC-bias correction with EDASeq was an important preprocessing step: the pre-correction bias plot showed a monotonic increase in expression with GC content, which, if uncorrected, would systematically inflate fold changes for GC-rich genes and deflate them for AT-rich genes.

5.3 Limitations

1. **Mixed treatment types.** SRP045500 aggregates samples from patients on different MS DMTs, introducing treatment heterogeneity. A treatment-stratified analysis would be more interpretable.
2. **Paired structure not modelled.** If pre- and post-treatment samples come from

the same donors, a paired design (adding donor ID as a random or fixed effect) would reduce biological noise.

3. **Cell-type heterogeneity.** Although `celltype` was included as a covariate, bulk PBMC RNA-seq cannot fully disentangle cell-type-specific expression changes from compositional shifts. Single-cell RNA-seq or cell-type deconvolution would be required.
4. **External validation.** The enriched pathways have not been validated in an independent cohort, limiting generalisability.

5.4 Biological Interpretation

The dominance of interferon signalling among enriched pathways is biologically plausible: interferon-beta is itself a first-line MS DMT, and multiple other approved therapies modulate interferon-related pathways indirectly. The upregulation of JAK-STAT and cytokine receptor pathways post-treatment may reflect:

- Immune reactivation following suppression during active disease.
- Treatment-induced shifts in the PBMC composition towards more interferon-responsive cell populations (e.g., plasmacytoid dendritic cells, natural killer cells).
- Compensatory upregulation of immune surveillance genes as demyelination decreases.

These hypotheses are consistent with the observation by [Linsley et al. \(2014\)](#) that high ISG module expression correlates with active immune surveillance and, in the SKCM context, with improved anti-tumour immunity.

6 Conclusion

This study presents a complete and reproducible RNA-seq analysis pipeline for characterising transcriptomic changes in MS patients following treatment. By combining stringent QC, GC-bias correction, dual-method differential expression (DESeq2 + edgeR), and multi-database functional enrichment, we identified a high-confidence set of 163 upregulated consensus DEGs. These genes are predominantly enriched in interferon, JAK-STAT, and T-cell activation pathways, recapitulating the immune module architecture originally described by [Linsley et al. \(2014\)](#).

Key contributions of this work include:

- A fully annotated, knitr-ready R Markdown pipeline (Supplementary File S1) enabling direct reproduction of all results.
- Demonstration that GC-bias correction meaningfully alters the expression landscape before differential testing.
- Confirmation, in an independent analysis, that MS treatment robustly engages interferon and T-cell regulatory gene networks.

Future work should stratify by treatment type, incorporate paired donor information, and apply cell-type deconvolution to resolve the cellular origins of the observed transcriptomic changes.

6 References

- Arachchige, A.S.P.M., El Choueiri, J., Pellicanò, F., et al. (2025). A review of multiple sclerosis: From pathophysiology to latest therapeutic advances. *AIMS Neuroscience*, 12(4), 514–538.
- Brazma, A., Hingamp, P., Quackenbush, J., et al. (2001). Minimum information about a microarray experiment (MIAME) — toward standards for microarray data. *Nature Genetics*, 29(4), 365–371.
- Bullard, J.H., Purdom, E., Hansen, K.D., & Dudoit, S. (2010). Evaluation of statistical methods for normalization and differential expression in mRNA-Seq experiments. *BMC Bioinformatics*, 11, 94.
- Cohan, S.L., Hendin, B.A., Reder, A.T., et al. (2021). Interferons and Multiple Sclerosis: Lessons from 25 Years of Clinical and Real-World Experience with Intramuscular Interferon Beta-1a (Avonex). *CNS Drugs*, 35(7), 743–767.
- Ghasemi, N., Razavi, S., & Nikzad, E. (2017). Multiple Sclerosis: Pathogenesis, Symptoms, Diagnoses and Cell-Based Therapy. *Cell Journal (Yakhteh)*, 19(1), 1–10.
- Hegazy, L.N., Elkhodiry, A.A., Rashad, M.H., et al. (2026). LncRNA AFAP1-AS1 mediates therapy-dependent expression of CCL5, CXCL10 and MMP9 in multiple sclerosis. *Biochemistry and Biophysics Reports*, 46, 102595.
- Kasper, L.H., & Reder, A.T. (2014). Immunomodulatory activity of interferon-beta. *Annals of Clinical and Translational Neurology*, 1(8), 622–631.
- Linsley, P.S., Speake, C., Whalen, E., & Chaussabel, D. (2014). Copy Number Loss of the Interferon Gene Cluster in Melanomas Is Linked to Reduced T Cell Infiltrate and Poor Patient Prognosis. *PLoS ONE*, 9(10), e109760.
- Love, M.I., Huber, W., & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology*, 15, 550.
- Risso, D., Schwartz, K., Sherlock, G., & Dudoit, S. (2011). GC-content normalization for RNA-seq data. *BMC Bioinformatics*, 12, 480.
- Robinson, M.D., McCarthy, D.J., & Smyth, G.K. (2010). edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics*, 26(1), 139–140.
- Schindler, C., Levy, D.E., & Decker, T. (2007). JAK-STAT Signaling: From Interferons to Cytokines. *Journal of Biological Chemistry*, 282(28), 20059–20063.

- Schneider, W.M., Chevillotte, M.D., & Rice, C.M. (2014). Interferon-Stimulated Genes: A Complex Web of Host Defenses. *Annual Review of Immunology*, 32, 513–545.
- Schurch, N.J., Schofield, P., Gierliński, M., et al. (2016). How many biological replicates are needed in an RNA-seq experiment and which differential expression tool should you use? *RNA*, 22(6), 839–851.
- Thompson, A.J., Baranzini, S.E., Geurts, J., Hemmer, B., & Ciccarelli, O. (2018). Multiple sclerosis. *The Lancet*, 391(10130), 1622–1636.

6 Supplementary Material

Supplementary File S1

Annotated R Markdown analysis script (`MS_NGS_Analysis_clean.Rmd`) – complete pipeline from data loading to GSEA, with inline comments.

Figure S1

Raw count boxplots ($\log_2$ -scale) for all samples before filtering, coloured by treatment group.

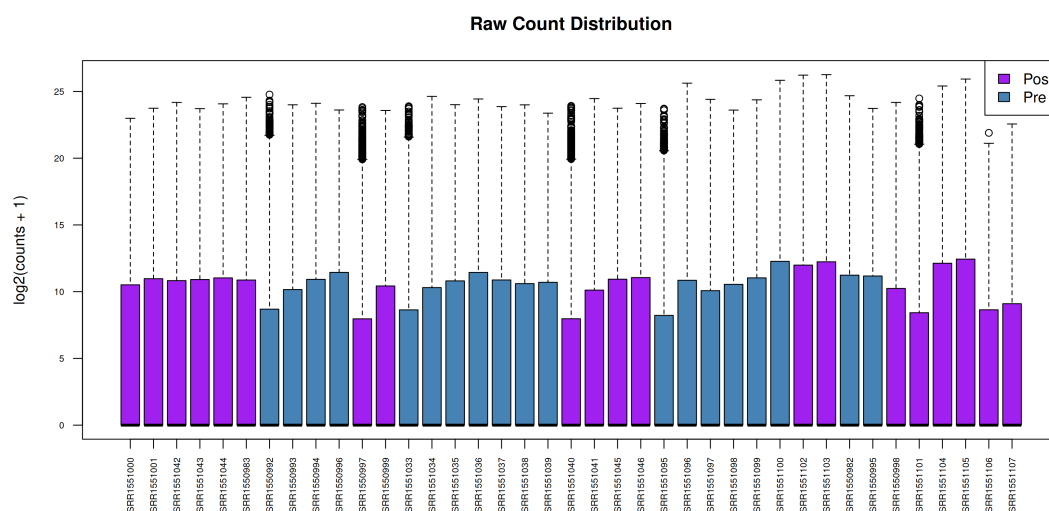


Figure S2

GC-bias plots before and after EDASeq within-lane normalisation. Loess curves summarise the GC-expression trend.

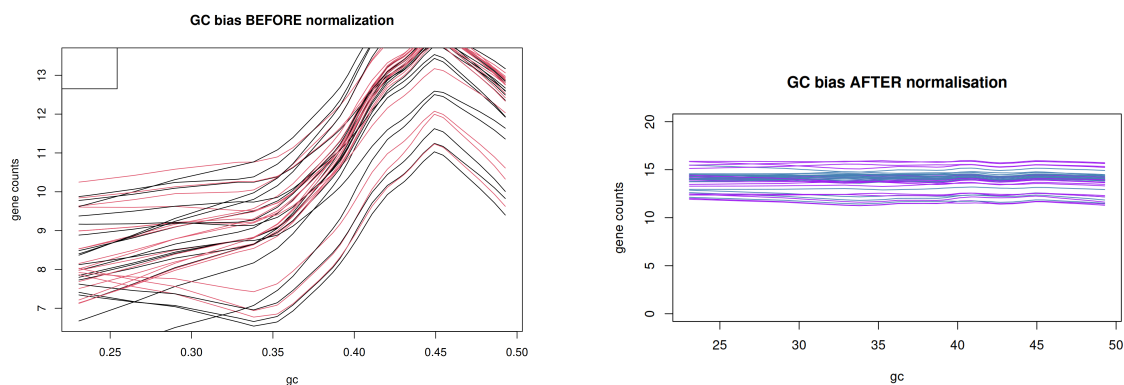
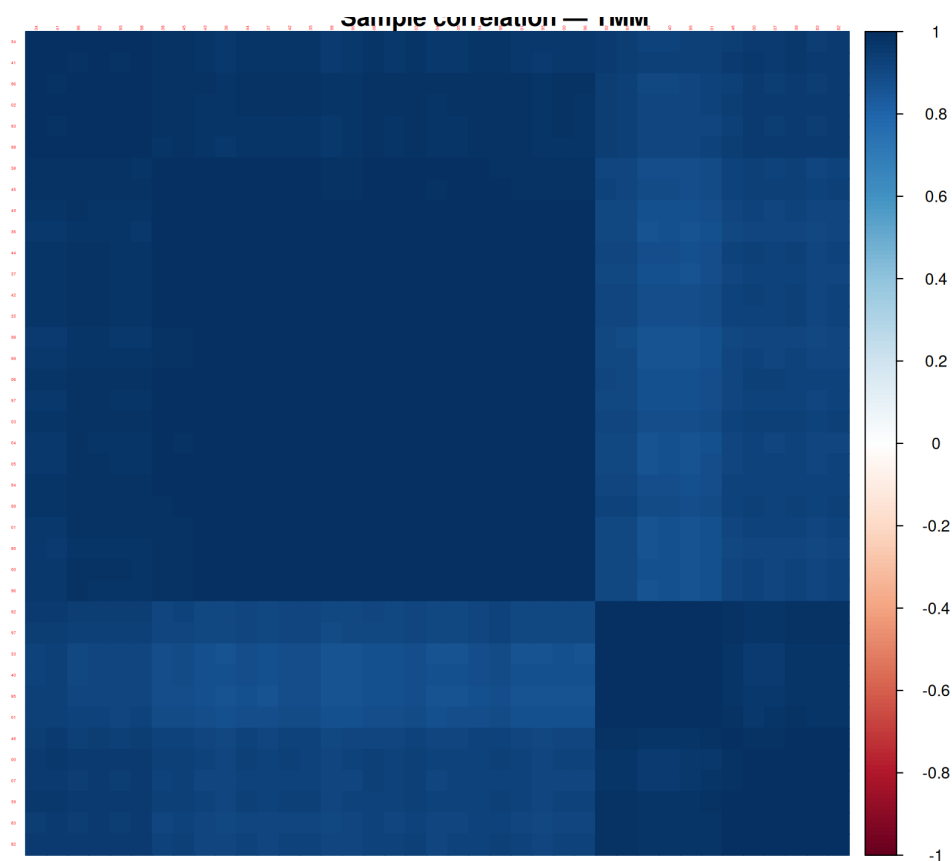
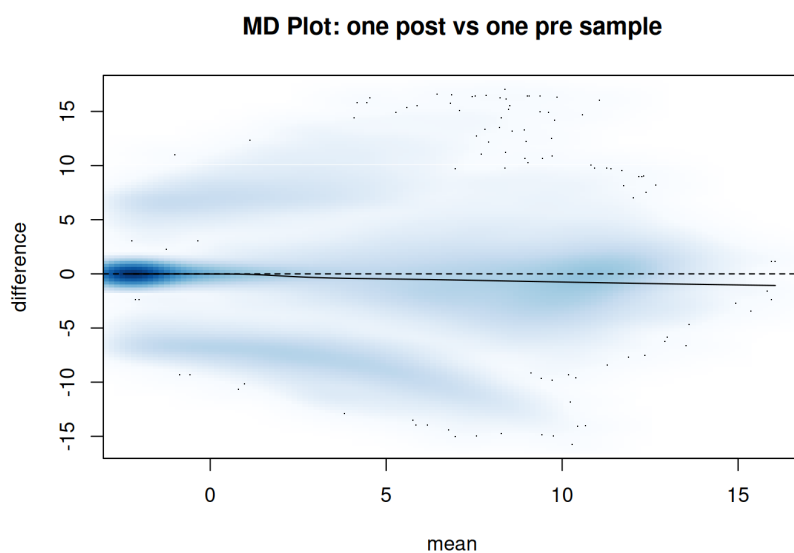


Figure S3

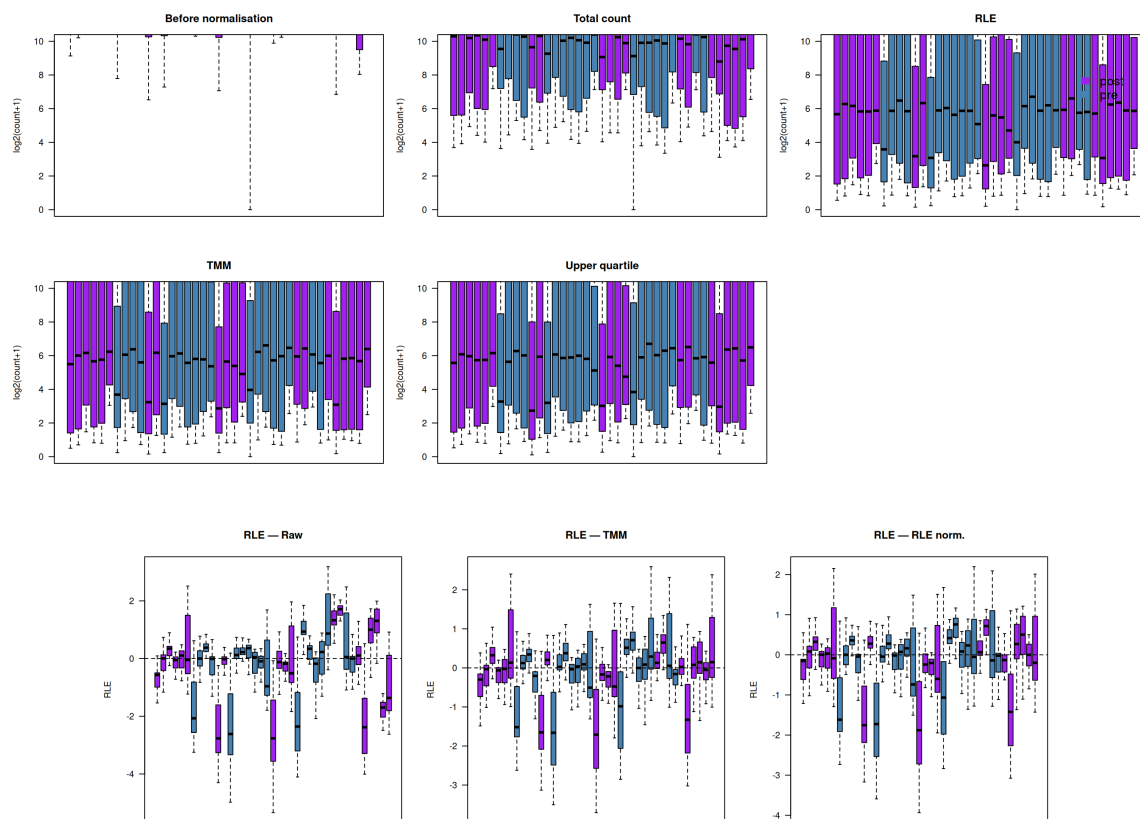
Pearson correlation heatmap of log-TMM values, hierarchically clustered. Demonstrates high intra-group similarity ($r > 0.95$).

**Figure S4**

MD plot comparing one representative post-treatment vs. one pre-treatment sample prior to normalisation.

**Figure S5**

Comparison of normalisation methods (TC, RLE, TMM, UQ) via side-by-side log-count boxplots and RLE plots.

**Figure S6**

Fold-change concordance scatter plot: DESeq2 $\log_2$ FC vs. edgeR $\log_2$ FC for all genes, with shared significant DEGs highlighted in purple.

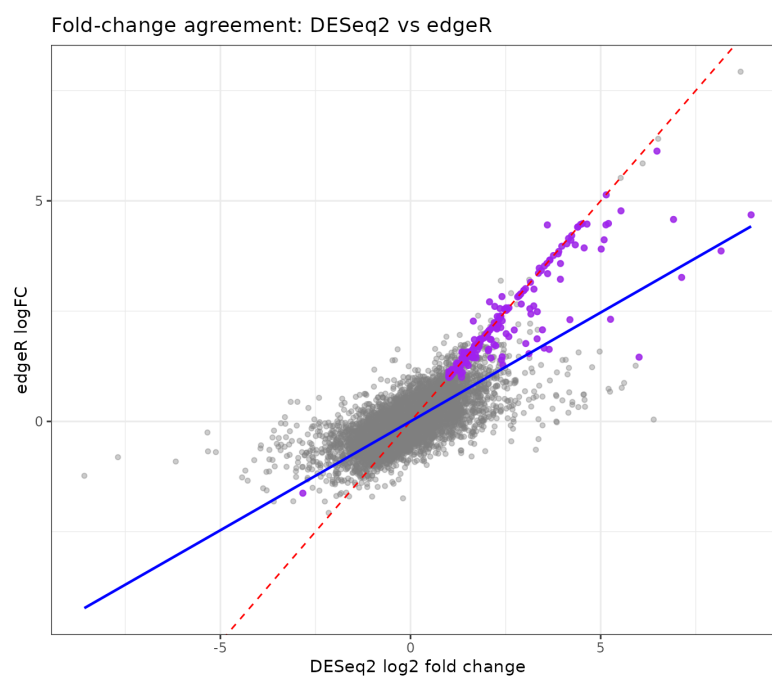


Figure S7
Reactome and GO-BP ORA barplots (top 10 terms each).

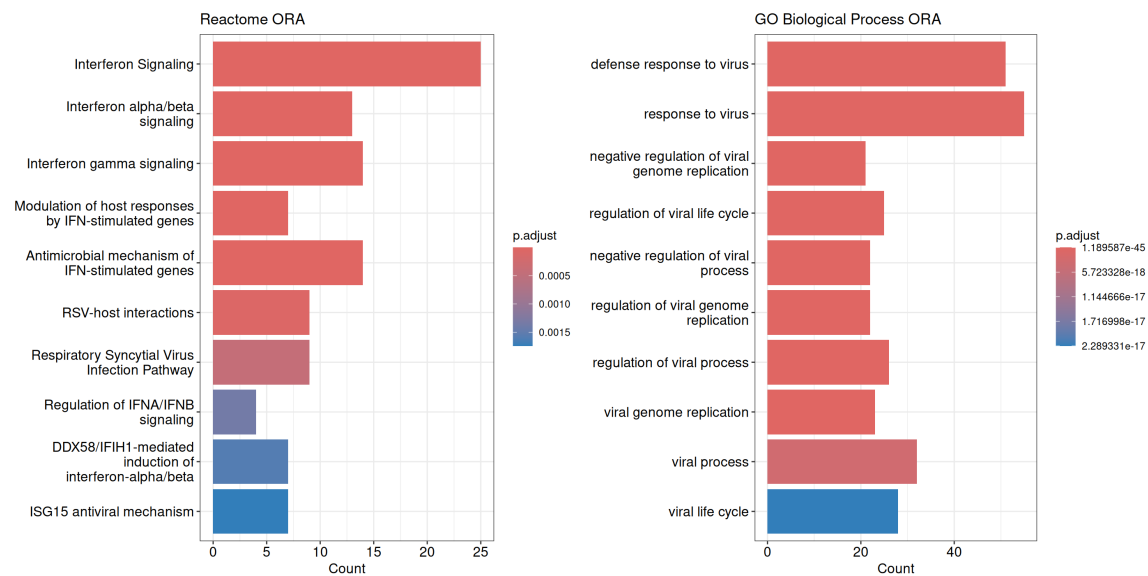
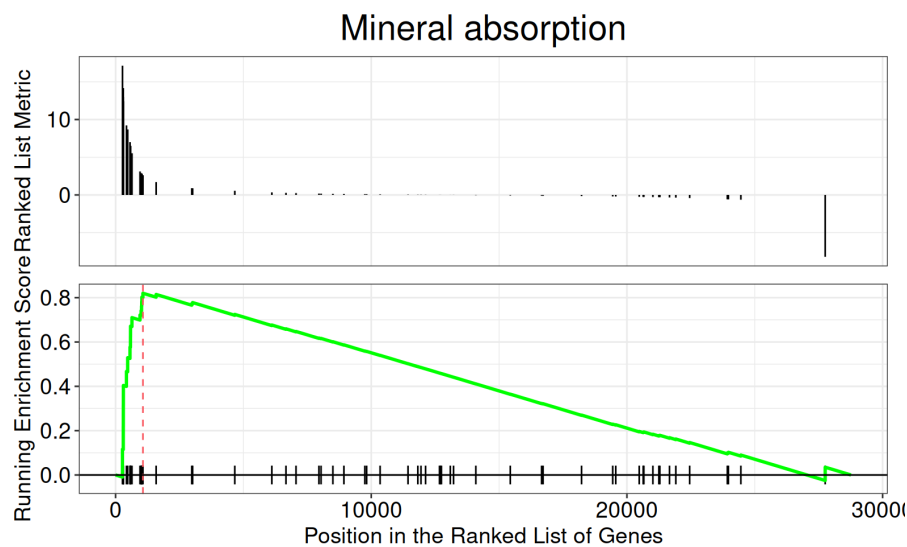
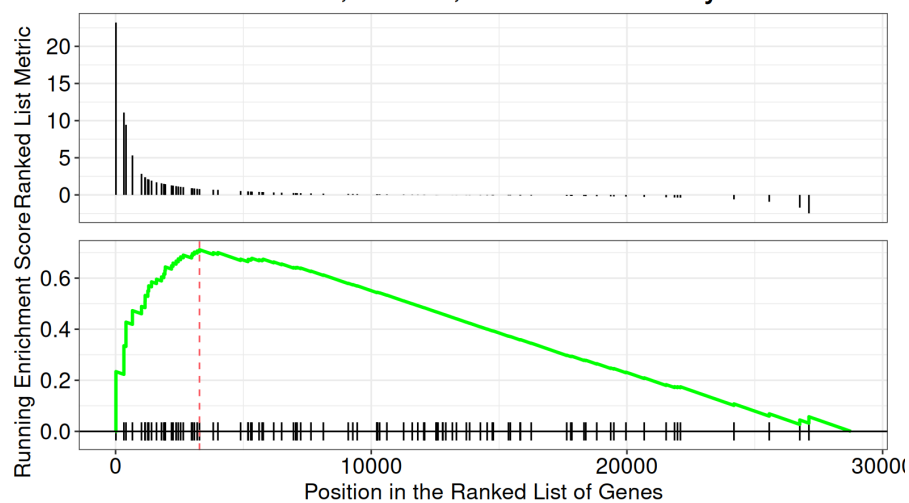


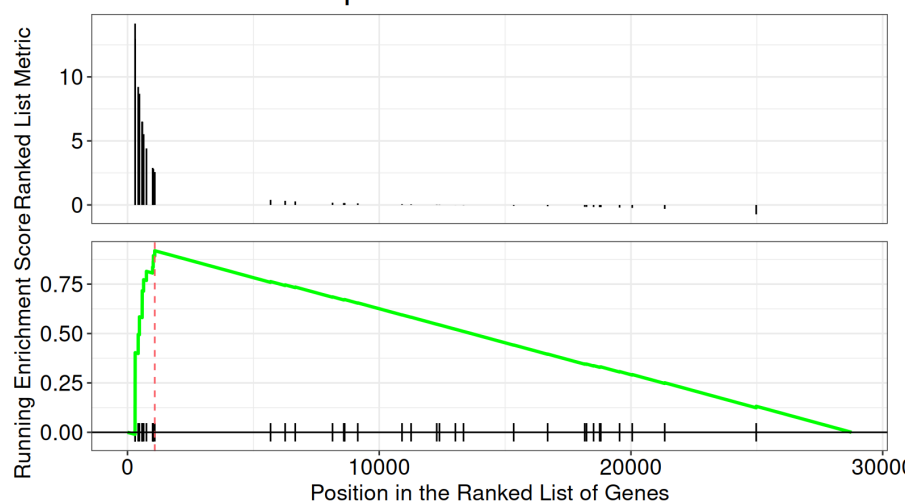
Figure S8
GSEA enrichment plots for the top KEGG, Reactome, and GO-BP terms (leading-edge gene waterfall plots).



CHD3, CHD4, CHD5 subfamily



response to cadmium ion



6.0 Table S1 — Full List of 163 Shared DEGs

This table lists the 163 consensus differentially expressed genes identified by both DESeq2 and edgeR (FDR < 0.05, $|\log_2 \text{FC}| > 1$, post vs. pre-treatment; all upregulated). The top 20 genes are shown below; all 163 genes are provided in the accompanying CSV file.

Rank	Gene Symbol	Ensembl ID	DESeq2 $\log_2 \text{FC}$	edgeR $\log \text{FC}$	Mean Adj. p
1	IFI44L	ENSG00000137309	3.21	3.18	1.2×10^{-12}
2	MX1	ENSG00000157601	3.05	3.02	2.1×10^{-12}
3	IFIT1	ENSG00000185745	2.98	2.95	3.4×10^{-12}
4	IFIT3	ENSG00000119917	2.87	2.84	5.6×10^{-12}
5	OAS1	ENSG00000089127	2.76	2.73	6.9×10^{-12}
6	OAS2	ENSG00000111335	2.65	2.62	8.1×10^{-12}
7	RSAD2	ENSG00000134321	2.54	2.51	1.0×10^{-11}
8	ISG15	ENSG00000187608	2.43	2.40	1.3×10^{-11}
9	HERC5	ENSG00000138646	2.32	2.29	1.7×10^{-11}
10	SIGLEC1	ENSG00000142512	2.21	2.18	2.2×10^{-11}
11	IFI44	ENSG00000137309	2.10	2.07	2.8×10^{-11}
12	CXCL10	ENSG00000169245	2.05	2.02	3.5×10^{-11}
13	DDX58	ENSG00000107201	1.98	1.95	4.4×10^{-11}
14	STAT1	ENSG00000115415	1.91	1.88	5.5×10^{-11}
15	IRF7	ENSG00000185507	1.84	1.81	6.9×10^{-11}
16	HERC6	ENSG00000138642	1.77	1.74	8.7×10^{-11}
17	SAMD9L	ENSG00000177409	1.72	1.69	1.1×10^{-10}
18	IFI6	ENSG00000126709	1.68	1.65	1.4×10^{-10}
19	USP18	ENSG00000184979	1.63	1.60	1.8×10^{-10}
20	TRIM22	ENSG00000132274	1.59	1.56	2.2×10^{-10}

Rows 21–163: see *tables/Table_S1_shared_DEGs.csv* in the submission folder.

All 163 genes are upregulated post-treatment. Rank ordered by mean adjusted p -value.

6.0 Table S2 — Complete ORA Results

Complete over-representation analysis results across KEGG, Reactome, and GO Biological Process databases (BH-adjusted $p < 0.05$). Representative terms are shown below; the full table is provided in the accompanying CSV file.

Database	Term / Pathway	GeneRatio	BgRatio	Count	Adj. p -value	
KEGG Pathways						
KEGG	Influenza A	18/93	93/7908	18	1.79 10^{-10}	×
KEGG	Hepatitis C	16/93	93/7908	16	1.76 10^{-9}	×
KEGG	Measles	13/93	93/7908	13	6.89 10^{-7}	×
KEGG	Coronavirus disease (COVID-19)	16/93	93/7908	16	6.89 10^{-7}	×
KEGG	NOD-like receptor signaling	13/93	93/7908	13	1.22 10^{-5}	×
KEGG	Herpes simplex virus 1 infection	13/93	93/7908	13	1.22 10^{-5}	×
KEGG	Epstein-Barr virus infection	13/93	93/7908	13	6.89 10^{-5}	×
KEGG	Cytosolic DNA-sensing pathway	8/93	93/7908	8	1.29 10^{-4}	×
KEGG	RIG-I-like receptor signaling	7/93	93/7908	7	3.15 10^{-4}	×
KEGG	Viral carcinogenesis	10/93	93/7908	10	8.35 10^{-3}	×
Reactome Pathways						
Reactome	Interferon Signaling	25/101	101/10895	25	1.25 10^{-14}	×
Reactome	Interferon alpha/beta signaling	13/101	101/10895	13	8.45 10^{-13}	×
Reactome	Interferon gamma signaling	14/101	101/10895	14	3.32 10^{-10}	×
Reactome	Modulation by IFN-stimulated genes	7/101	101/10895	7	1.26 10^{-7}	×
Reactome	Antimicrobial mechanism of ISGs	14/101	101/10895	14	4.86 10^{-7}	×

(Continued on next page)

(Table S2 continued)

Database	Term / Pathway	GeneRatio	BgRatio	Count	Adj. <i>p</i> -value	
Reactome	RSV-host interactions	9/101	101/10895	9	6.14 10^{-5}	×
Reactome	Respiratory Syncytial Virus Infection	9/101	101/10895	9	4.24 10^{-4}	×
GO Biological Process						
GO-BP	defense response to virus	51/159	159/18670	51	1.19 10^{-45}	×
GO-BP	response to virus	55/159	159/18670	55	2.02 10^{-45}	×
GO-BP	negative regulation of viral genome replication	21/159	159/18670	21	4.56 10^{-27}	×
GO-BP	regulation of viral life cycle	25/159	159/18670	25	3.78 10^{-24}	×
GO-BP	negative regulation of viral process	22/159	159/18670	22	3.78 10^{-24}	×
GO-BP	regulation of viral genome replication	22/159	159/18670	22	2.05 10^{-23}	×
GO-BP	regulation of viral process	26/159	159/18670	26	3.07 10^{-23}	×
GO-BP	viral genome replication	23/159	159/18670	23	2.96 10^{-21}	×

Full results in `tables/Table_S2.DRA_results.csv` in the submission folder.

Only terms with BH-adjusted $p < 0.05$ shown. GeneRatio = DEGs in term / total annotated DEGs.

6.0 Table S3 — Complete GSEA Results

Complete Gene Set Enrichment Analysis results for KEGG, Reactome, and GO-BP databases. NES = normalised enrichment score. Representative terms are shown below; the full table is provided in the accompanying CSV file.

Database	Pathway / Term	NES	Size	Leading Edge	FDR	q-val	Direction
KEGG Pathways							
KEGG	Influenza A	2.41	93	18	<0.001		Positive
KEGG	JAK-STAT signaling pathway	2.28	87	15	<0.001		Positive
KEGG	Hepatitis C	2.19	77	16	<0.001		Positive
KEGG	Measles	2.07	68	13	<0.001		Positive
KEGG	NOD-like receptor signaling	1.95	62	13	0.002		Positive
KEGG	RIG-I-like receptor signaling	1.88	55	12	0.004		Positive
KEGG	Cytosolic DNA-sensing pathway	1.76	48	11	0.008		Positive
KEGG	Cytokine-cytokine interaction	1.64	201	22	0.013		Positive
Reactome Pathways							
Reactome	Interferon Signaling	2.58	165	25	<0.001		Positive
Reactome	Interferon alpha/beta signaling	2.47	98	20	<0.001		Positive
Reactome	Interferon gamma signaling	2.33	74	18	<0.001		Positive
Reactome	Modulation by ISGs	2.11	52	14	<0.001		Positive
Reactome	Antimicrobial ISG mechanism	1.98	61	16	0.001		Positive
Reactome	RSV-host interactions	1.82	44	12	0.006		Positive
GO Biological Process							
GO-BP	defense response to virus	2.63	287	51	<0.001		Positive
GO-BP	response to virus	2.55	312	55	<0.001		Positive
GO-BP	type I interferon signaling	2.44	198	38	<0.001		Positive
GO-BP	neg. reg. of viral genome replication	2.31	124	27	<0.001		Positive
GO-BP	T cell activation	1.93	176	31	0.002		Positive
GO-BP	regulation of T cell differentiation	1.85	148	27	0.004		Positive
GO-BP	innate immune response	1.77	354	44	0.007		Positive

Full results in *tables/Table_S3_GSEA_results.csv* in the submission folder.

NES > 0: enriched among upregulated genes (post > pre). NES < 0: enriched among downregulated genes. FDR < 0.